

# Finerenone with Empagliflozin in Chronic Kidney Disease and Type 2 Diabetes

Critical Appraisal of the CONFIDENCE Trial

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**RCT**



# Trial Identity and Citation

## TRIAL NAME & CITATION

**CONFIDENCE Trial:** Finerenone with Empagliflozin in Chronic Kidney Disease and Type 2 Diabetes

**Citation:** N Engl J Med 2025;393:533-43

## FUNDING & RESEARCH GROUPS

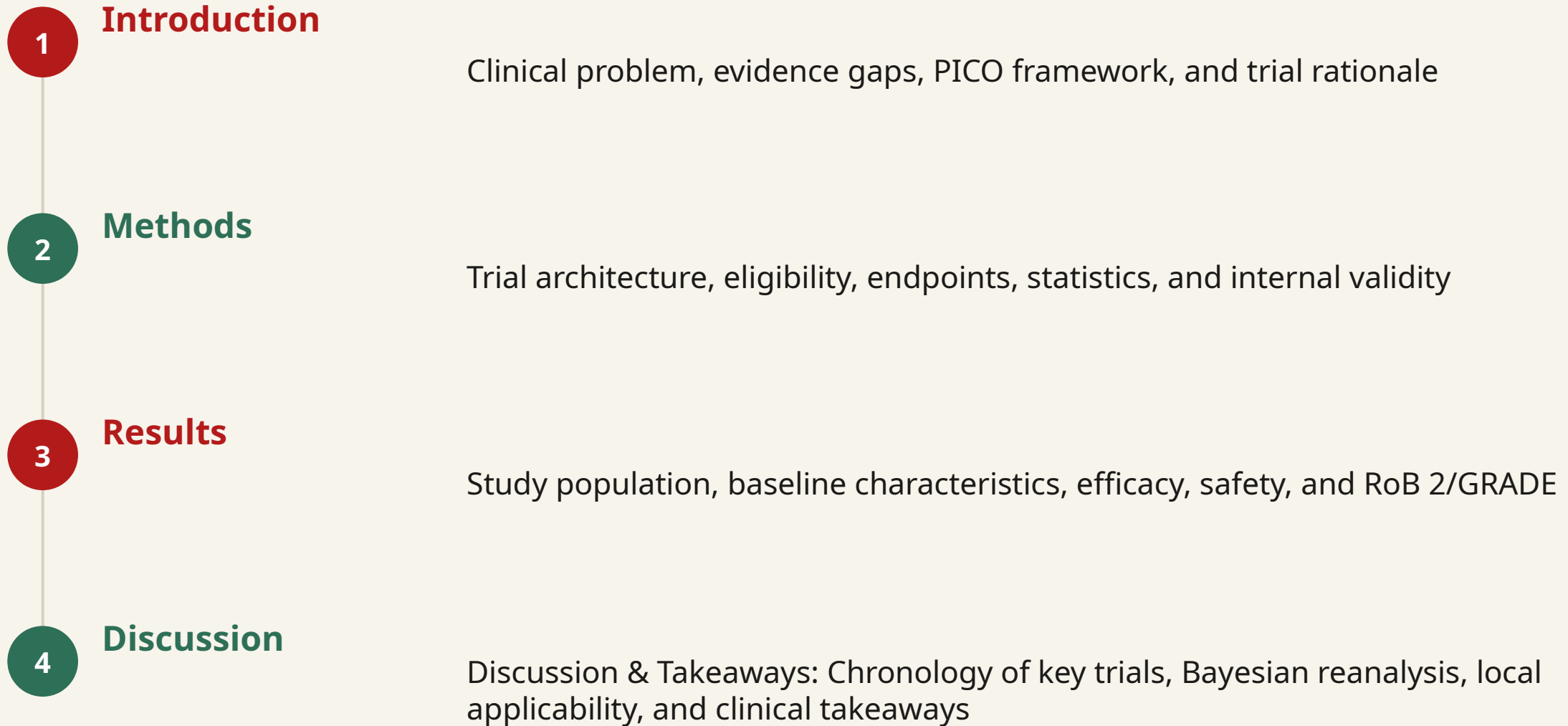
- Full Citation: Agarwal R, Green JB, Heerspink HJL, et al. Finerenone with Empagliflozin in Chronic Kidney Disease and Type 2 Diabetes. N Engl J Med 2025;393:533-543. doi:10.1056/NEJMoa2410659
- Trial Name: CONFIDENCE (Combination Effect of Finerenone and Empagliflozin).
- Trial Registration: ClinicalTrials.gov number, NCT05254002.

## TRIAL REGISTRATION & SPONSORS

- Funding: Supported and funded by Bayer AG, Berlin, Germany.
- Sponsors: Bayer participated in the study design, analysis, and manuscript preparation assistance.
- Oversight: Approved by site IRBs; safety overseen by an independent data and safety monitoring committee.

# Structure of This Appraisal

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# INTRODUCTION

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Background, Clinical Need, and Trial Rationale

# Why This Paper Matters

**Clinical problem:** Clinical Problem: Patients with chronic kidney disease (CKD) and type 2 diabetes face high risk of progressive kidney disease and cardiovascular events despite standard RAS inhibition.

**Evidence gap:** Evidence Gap: SGLT2 inhibitors and finerenone independently protect organs in CKD. Although expert consensus suggested additive benefit, simultaneous dual initiation had not been tested in a rigorous double-blind RCT.

**Research question:** Research Question: Does the simultaneous initiation of finerenone and empagliflozin lead to a greater reduction in albuminuria than either monotherapy alone, and is this strategy safe?

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# Clinical Question & PICO Framework

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- P** — Adults with type 2 diabetes and chronic kidney disease (eGFR 30 to 90 ml/min/1.73m<sup>2</sup>, UACR 100 to 5000 mg/g) receiving stable maximum labeled dose of an ACE inhibitor or ARB.
- I** — Combined initiation of finerenone (10 or 20 mg/day) and empagliflozin (10 mg/day).
- C** — Monotherapy of either finerenone alone (with empagliflozin placebo) OR empagliflozin alone (with finerenone placebo).
- O** — Primary outcome: relative change in log-transformed mean urinary albumin-to-creatinine ratio (UACR) from baseline to 180 days. Safety outcomes: hyperkalemia, acute kidney injury, eGFR decline, and adverse events.

# METHODS

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Study Population, Eligibility, Outcomes, and Statistics

# Study Design & Methodology

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## TRIAL ARCHITECTURE

- Design Type: Double-blind, randomized, active-controlled, parallel-group, phase 2 trial.
- Randomization: 1:1:1 ratio using an interactive Web-response system, stratified by baseline eGFR (<60 vs ≥60 ml/min/1.73m<sup>2</sup>) and UACR (≤850 vs >850 mg/g).
- Setting: International multicenter trial (multiple countries and regions).

## BIAS CONTROLS

- Blinding Status: Double-blind; use of matching placebos (finerenone-matching and empagliflozin-matching placebos) for all participants to maintain strict masking.
- Trial Oversight: Steering committee, independent data and safety monitoring committee, and independent statistical confirmation of data.

# Eligibility Criteria

## KEY INCLUSION CRITERIA

- Age & Disease: Adults with type 2 diabetes and glycated hemoglobin (HbA1c) < 11.0%.
- Renal Function: Estimated GFR between 30 and 90 ml/min/1.73m<sup>2</sup>.
- Albuminuria: UACR between 100 and 5000 mg/g (based on 3 morning urines).

## KEY EXCLUSION CRITERIA

- Heart Failure: Excluded patients with chronic heart failure, reduced ejection fraction, and persistent symptoms.
- Electrolytes: Excluded patients with a screening serum potassium level above 4.8 mmol/L.
- Acute Events: Excluded patients with myocardial infarction, stroke, or hospitalization for heart failure within 90 days.

***Key Prerequisite: Participants must have been receiving a stable maximum labeled dose of an ACE inhibitor or ARB for more than 1 month at screening.***

# Trial Endpoints & Definitions

## PRIMARY EFFICACY

- Relative change in the log-transformed mean urinary albumin-to-creatinine ratio (UACR) from baseline to 180 days.
- UACR assessed on consecutive morning urine samples analyzed at a central laboratory.

## SECONDARY EFFICACY

- Proportions of patients achieving >30%, >40%, or >50% reductions in UACR at day 180.
- Reversibility: Relative change in UACR 30 days after stopping study drug (day 210 vs day 180) to assess washout return.

## SECONDARY SAFETY

- Acute eGFR drop from baseline to day 30 and its reversibility at day 210.
- Hyperkalemia: incidence of serum potassium >5.5 mmol/L and >6.0 mmol/L.
- Clinical events: acute kidney injury (AKI), symptomatic hypotension, and ketoacidosis.

# Statistical Plan & Power

**807**

target enrollment

**80%**

efficacy power

**0.05**

two-sided alpha

**15%**

assumed trial dropout

- Power Analysis: Calculated that 807 participants would provide 80% power to detect a 20% greater reduction in UACR with combination than with either monotherapy, assuming 15% dropout.
- Hypothesis Testing: Applied a two-sample t-test on log-transformed changes. Multiple testing adjusted using the Holm-Bonferroni method for two primary comparisons.
- Analysis Populations: Efficacy analyzed in the full analysis population (randomized, received  $\geq 1$  dose, and no critical GCP violations). Safety analyzed in all treated patients.
- Missing Data: Handled using multiple imputation under the missing-at-random (MAR) assumption, with a tipping-point analysis assuming missing-not-at-random (MNAR).

# Protocol Adherence and Washout

**818**

underwent randomization

**800**

included in efficacy (FAP)

**798**

included in safety

**746**

completed 30-day washout

- Screening: 1664 participants screened, 818 randomized from June 2022 to August 2024. Database locked April 2025.
- Exclusions: 4 randomized in error (never dosed) and 14 from one site excluded for Good Clinical Practice violations unrelated to study conduct.
- Washout: Regimen stopped at 180 days. A total of 746 participants completed the 30-day follow-up (day 210) to assess reversibility.
- FAP Breakdown: 269 assigned to combination, 264 to finerenone alone, and 267 to empagliflozin alone.

# Methods Critical Appraisal: Internal Validity

## METHODOLOGICAL STRENGTHS

- Triple Masking: Double-blind, double-dummy setup successfully blinded patients, treating physicians, and outcome assessors.
- Stratification: Randomization was stratified by baseline eGFR and UACR, ensuring excellent prognostic balance across groups.
- Missing Data: High study completion rate; multiple imputation and tipping-point analyses confirmed the statistical findings.

## LIMITATIONS & CONCERNS

- Surrogate Endpoint: The trial used UACR change rather than hard clinical kidney (ESKD, death) or CV endpoints.
- Bayer Sponsorship: Fully funded and coordinated by the developer of finerenone (Bayer), representing a potential source of funding bias.

***Internal Validity Verdict: Extremely high internal validity due to double-blind double-dummy design, though clinically limited by reliance on a surrogate endpoint (UACR).***

# RESULTS

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Enrollment, Baseline Characteristics, and Efficacy Outcome Data

# Table 1. Demographic and Clinical Characteristics at Baseline

| Table 1. Demographic and Clinical Characteristics of the Participants at Baseline.* |                                         |                      |                         |                 |
|-------------------------------------------------------------------------------------|-----------------------------------------|----------------------|-------------------------|-----------------|
| Characteristic                                                                      | Finerenone plus Empagliflozin (N = 269) | Finerenone (N = 264) | Empagliflozin (N = 267) | Total (N = 800) |
| Age — yr                                                                            | 67.7±10.0                               | 65.5±10.7            | 66.2±10.1               | 66.5±10.3       |
| Female sex — no. (%)                                                                | 67 (24.9)                               | 61 (23.1)            | 70 (26.2)               | 198 (24.8)      |
| Race or ethnic group — no. (%)†                                                     |                                         |                      |                         |                 |
| White                                                                               | 130 (48.3)                              | 105 (39.8)           | 116 (43.4)              | 351 (43.9)      |
| Asian                                                                               | 114 (42.4)                              | 132 (50.0)           | 125 (46.8)              | 371 (46.4)      |
| Asian Indian                                                                        | 34 (12.6)                               | 54 (20.5)            | 45 (16.9)               | 133 (16.6)      |
| Black                                                                               | 22 (8.2)                                | 21 (8.0)             | 24 (9.0)                | 67 (8.4)        |
| Other                                                                               | 2 (0.7)                                 | 2 (0.8)              | 1 (0.4)                 | 5 (0.6)         |
| Geographic region — no. (%)                                                         |                                         |                      |                         |                 |
| Europe                                                                              | 81 (30.1)                               | 62 (23.5)            | 72 (27.0)               | 215 (26.9)      |
| North America                                                                       | 75 (27.9)                               | 79 (29.9)            | 71 (26.6)               | 225 (28.1)      |
| Asia                                                                                | 113 (42.0)                              | 123 (46.6)           | 124 (46.4)              | 360 (45.0)      |
| Glycated hemoglobin — %‡                                                            | 7.3±1.2                                 | 7.3±1.2              | 7.3±1.2                 | 7.3±1.2         |
| Body weight — kg                                                                    | 83.0±21.7                               | 81.9±20.3            | 80.8±19.5               | 81.9±20.5       |
| Body-mass index§                                                                    | 29.8±6.7                                | 29.1±5.7             | 29.0±5.7                | 29.3±6.1        |
| eGFR — ml/min/1.73 m²¶                                                              | 53.9±16.9                               | 54.3±17.8            | 54.2±16.6               | 54.2±17.1       |
| Median urinary albumin-to-creatinine ratio (IQR)                                    | 574 (274–999)                           | 578 (292–1289)       | 583 (301–1114)          | 579 (292–1092)  |
| Urinary albumin-to-creatinine ratio category — no. (%)**                            |                                         |                      |                         |                 |
| ≤850                                                                                | 184 (68.4)                              | 166 (62.9)           | 170 (63.7)              | 520 (65.0)      |
| >850                                                                                | 81 (30.1)                               | 92 (34.8)            | 91 (34.1)               | 264 (33.0)      |
| Systolic blood pressure — mm Hg                                                     | 135.0±13.7                              | 135.1±13.7           | 135.4±12.6              | 135.2±13.3      |
| Serum potassium — mmol/liter††                                                      | 4.4±0.4                                 | 4.5±0.5              | 4.5±0.4                 | 4.5±0.4         |
| History of atherosclerotic cardiovascular disease — no. (%)‡‡                       | 75 (27.9)                               | 71 (26.9)            | 78 (29.2)               | 224 (28.0)      |
| Concomitant medications — no. (%)                                                   |                                         |                      |                         |                 |
| ACE inhibitor or ARB§§                                                              | 267 (99.3)                              | 260 (98.5)           | 260 (97.4)              | 787 (98.4)      |
| Statins                                                                             | 209 (77.7)                              | 197 (74.6)           | 191 (71.5)              | 597 (74.6)      |
| Diuretics                                                                           | 108 (40.1)                              | 97 (36.7)            | 84 (31.5)               | 289 (36.1)      |
| Antihyperglycemic agents — no. (%)                                                  |                                         |                      |                         |                 |
| Insulin                                                                             | 108 (40.1)                              | 96 (36.4)            | 113 (42.3)              | 317 (39.6)      |
| GLP-1 RA                                                                            | 68 (25.3)                               | 52 (19.7)            | 62 (23.2)               | 182 (22.8)      |

\* Plus-minus values are means ±SD. Percentages may not total 100 because of rounding. GLP-1 RA denotes glucagon-like peptide-1 receptor agonist, and IQR interquartile range.

† Race or ethnic group was reported by the participant. “Other” includes American Indian, Alaska Native, Native Hawaiian, or other Pacific Islander.

‡ Data on the glycated hemoglobin level were missing for 19 participants (9 [3.3%] in the combination-therapy [finerenone plus empagliflozin] group, 7 [2.7%] in the finerenone group, and 3 [1.1%] in the empagliflozin group).

§ The body-mass index is the weight in kilograms divided by the square of the height in meters.

¶ The estimated glomerular filtration rate (eGFR) was calculated with the use of the Chronic Kidney Disease Epidemiology Collaboration equation,<sup>8</sup> which was modified for the Japanese participants.<sup>9</sup> Data were missing for 4 participants (2 [0.8%] in the finerenone group and 2 [0.7%] in the empagliflozin group).

|| The urinary albumin-to-creatinine ratio was calculated with albumin measured in milligrams and creatinine measured in grams. Data were missing for 16 participants (4 [1.5%] in the combination-therapy group, 6 [2.3%] in the finerenone group, and 6 [2.2%] in the empagliflozin group).

\*\* Albuminuria was defined as a urinary albumin-to-creatinine ratio between 100 and 5000, as determined according to the mean value of the first urine specimens on three consecutive mornings.

†† Data on the serum potassium level were missing for 5 participants (2 [0.7%] in the combination-therapy group, 1 [0.4%] in the finerenone group, and 2 [0.7%] in the empagliflozin group).

‡‡ A history of atherosclerotic cardiovascular disease was based on a group of preferred terms for coronary artery disease, cerebral infarction, and stroke (a transient ischemic attack alone is not sufficient), as well as for peripheral artery disease and carotid revascularization, in the *Medical Dictionary for Regulatory Activities*, version 27.1.

§§ According to the protocol, all the participants were required to be taking an angiotensin-converting-enzyme (ACE) inhibitor or angiotensin-receptor blocker (ARB) at the clinically maximum tolerated dose.

Table 1. Baseline characteristics of the participants across the three treatment arms.

# Baseline Comparability Critique

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- Key Balances: Mean age 66.5 years, 24.8% female, mean eGFR 54.2 ml/min/1.73m<sup>2</sup>, and median baseline UACR 579 mg/g (IQR 292–1092).
- Background Therapy: ACE inhibitors or ARBs used in 98.4% of participants at baseline. Statins used in 74.6%, and diuretics in 36.1%.
- Cardiovascular Comorbidity: 28.0% had a history of atherosclerotic cardiovascular disease, which was balanced among the arms.
- Imbalances: No clinically or statistically significant baseline differences existed, ruling out confounding by baseline characteristics.

***Verdict: Randomization succeeded. Baseline characteristics, age (66.5 years), glycated hemoglobin (7.3%), eGFR (54.2 ml/min/1.73m<sup>2</sup>), and median UACR (579 mg/g) were highly balanced across all three arms.***

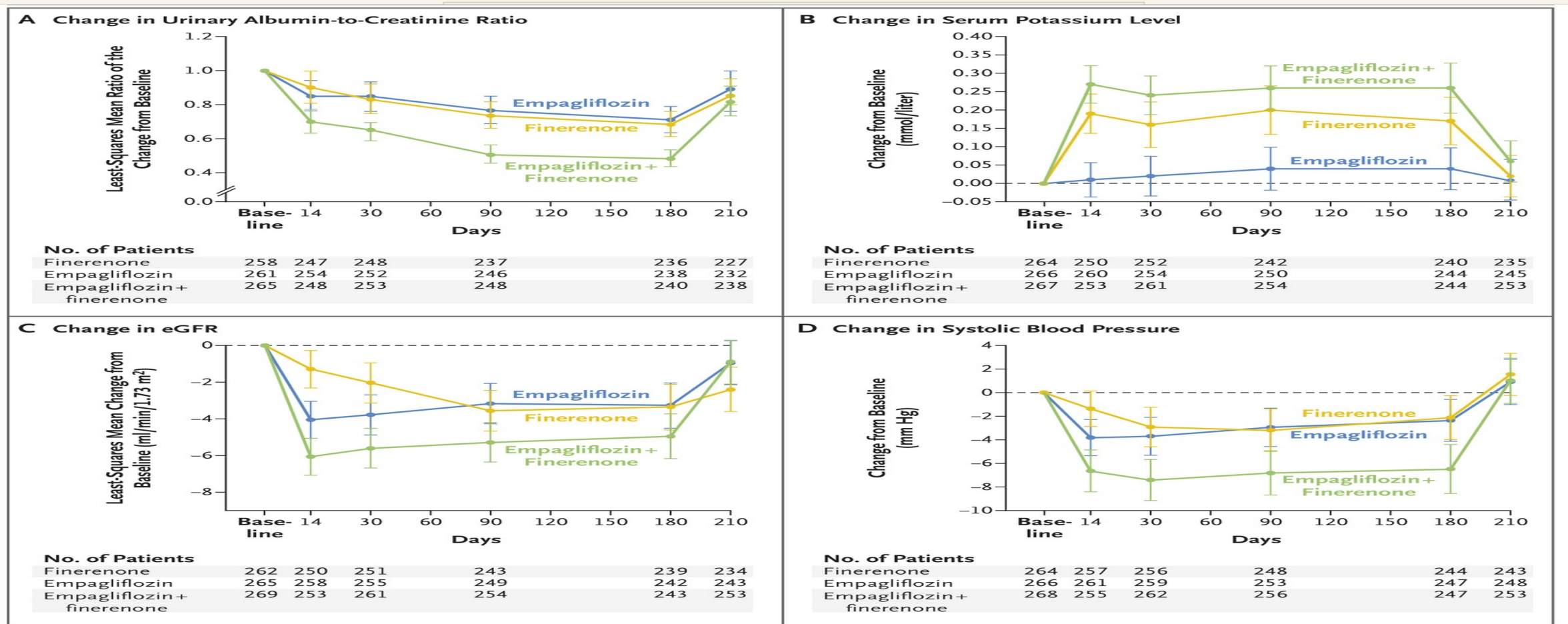
Table 2. Primary and Secondary Efficacy Outcomes

| Table 2. Primary and Secondary Outcomes for the Change in Urinary Albumin-to-Creatinine Ratio. |                                            |                                       |                                      |                                                        |                                      |                                                        |
|------------------------------------------------------------------------------------------------|--------------------------------------------|---------------------------------------|--------------------------------------|--------------------------------------------------------|--------------------------------------|--------------------------------------------------------|
| Variable                                                                                       | Finerenone plus Empagliflozin<br>(N = 269) |                                       | Finerenone<br>(N = 264)              |                                                        | Empagliflozin<br>(N = 267)           |                                                        |
|                                                                                                | No. of<br>Participants                     | Least-Squares Mean<br>Ratio (95% CI)* | No. of<br>Participants               | Least-Squares Mean<br>Ratio or Odds Ratio<br>(95% CI)* | No. of<br>Participants               | Least-Squares Mean<br>Ratio or Odds Ratio<br>(95% CI)* |
| Change in urinary albumin-to-creatinine ratio                                                  |                                            |                                       |                                      |                                                        |                                      |                                                        |
| Primary outcome: change from baseline to day 180                                               | 240                                        | 0.48 (0.44–0.54)                      | 236                                  | 0.68 (0.61–0.76)                                       | 238                                  | 0.71 (0.64–0.79)                                       |
| Difference in change, combination vs. monotherapy                                              |                                            |                                       |                                      | 0.71 (0.61–0.82)†                                      |                                      | 0.68 (0.59–0.79)†                                      |
| Change from baseline to day 210                                                                | 238                                        | 0.82 (0.73–0.91)                      | 227                                  | 0.85 (0.76–0.96)                                       | 232                                  | 0.90 (0.80–1.00)                                       |
| Difference in change, combination vs. monotherapy                                              |                                            |                                       |                                      | 0.96 (0.82–1.12)                                       |                                      | 0.91 (0.79–1.06)                                       |
| Change from day 180 to day 210                                                                 | 239                                        | 1.63 (1.49–1.78)                      | 234                                  | 1.45 (1.32–1.59)                                       | 238                                  | 1.44 (1.32–1.58)                                       |
| Difference in change, combination vs. monotherapy                                              |                                            |                                       |                                      | 1.13 (1.00–1.27)                                       |                                      | 1.13 (1.00–1.28)                                       |
|                                                                                                | No. with Reduction/<br>Total No. (%)       |                                       | No. with Reduction/<br>Total No. (%) |                                                        | No. with Reduction/<br>Total No. (%) |                                                        |
| Reduction in urinary albumin-to-creatinine ratio according to prespecified thresholds          |                                            |                                       |                                      |                                                        |                                      |                                                        |
| Reduction from baseline to day 180                                                             |                                            |                                       |                                      |                                                        |                                      |                                                        |
| >30%                                                                                           | 168/240 (70.0)                             |                                       | 123/236 (52.1)                       |                                                        | 123/238 (51.7)                       |                                                        |
| >40%                                                                                           | 154/240 (64.2)                             |                                       | 104/236 (44.1)                       |                                                        | 103/238 (43.3)                       |                                                        |
| >50%                                                                                           | 131/240 (54.6)                             |                                       | 84/236 (35.6)                        |                                                        | 76/238 (31.9)                        |                                                        |
| Reduction from baseline to day 210                                                             |                                            |                                       |                                      |                                                        |                                      |                                                        |
| >30%                                                                                           | 111/238 (46.6)                             |                                       | 78/227 (34.4)                        |                                                        | 92/232 (39.7)                        |                                                        |
| >40%                                                                                           | 85/238 (35.7)                              |                                       | 62/227 (27.3)                        |                                                        | 69/232 (29.7)                        |                                                        |
| >50%                                                                                           | 65/238 (27.3)                              |                                       | 50/227 (22.0)                        |                                                        | 47/232 (20.3)                        |                                                        |

\* Least-squares mean ratios are shown for the changes in the urinary albumin-to-creatinine ratio, and odds ratios (combination vs. monotherapy) are shown for the reductions in the urinary albumin-to-creatinine ratio according to prespecified thresholds. The widths of the 95% confidence intervals for outcomes other than the primary outcome have not been adjusted for multiplicity and should not be used to test for efficacy.

† P<0.001.

# Figure 1. Efficacy and Safety Biomarkers Over Time



**Figure 1. Urinary Albumin-to-Creatinine Ratio, Serum Potassium Level, Estimated Glomerular Filtration Rate, and Systolic Blood Pressure over Time.**

Panel A shows the least-squares mean ratio of the change from baseline in the urinary albumin-to-creatinine ratio in the full analysis population (all the participants who had undergone randomization, with the exclusion of those who had not received at least one dose of a trial drug and those with critical violations of Good Clinical Practice guidelines). Multiple imputation was used to account for missing data, with the assumption that data were missing at random. Panel B shows the mean serum potassium level at each visit in the safety analysis population (all the participants who had undergone randomization and received at least one dose of a trial medication). Panel C shows the least-squares mean change in the estimated glomerular filtration rate (eGFR) in the safety analysis population. The eGFR was calculated with the use of the Chronic Kidney Disease Epidemiology Collaboration equation,<sup>8</sup> which was modified for the Japanese participants,<sup>9</sup> and the analysis was performed with the use of a mixed model for repeated measures. Panel D shows the mean systolic blood pressure at each visit in the safety analysis population. Summary statistics for systolic blood pressure were derived from the mean value of all assessments (i.e., three assessments per visit per participant).

Figure 1. Changes in UACR (A), Serum Potassium (B), eGFR (C), and Systolic Blood Pressure (D) over 180 days.

# Table 3. Adverse Events and Safety Assessments

| Table 3. Adverse Events and Safety Assessments after Treatment Initiation (Safety Analysis Population).* |                                               |                         |                            |                    |
|----------------------------------------------------------------------------------------------------------|-----------------------------------------------|-------------------------|----------------------------|--------------------|
| Event or Assessment                                                                                      | Finerenone plus<br>Empagliflozin<br>(N = 268) | Finerenone<br>(N = 264) | Empagliflozin<br>(N = 266) | Total<br>(N = 798) |
|                                                                                                          | number (percent)                              |                         |                            |                    |
| Investigator-reported adverse events†                                                                    |                                               |                         |                            |                    |
| Any adverse event                                                                                        | 144 (53.7)                                    | 136 (51.5)              | 135 (50.8)                 | 415 (52.0)         |
| Adverse event leading to treatment discontinuation                                                       | 12 (4.5)                                      | 9 (3.4)                 | 9 (3.4)                    | 30 (3.8)           |
| Any serious adverse event                                                                                | 19 (7.1)                                      | 16 (6.1)                | 17 (6.4)                   | 52 (6.5)           |
| Serious adverse event leading to treatment discontinuation                                               | 3 (1.1)                                       | 3 (1.1)                 | 2 (0.8)                    | 8 (1.0)            |
| Adverse event with death as the outcome                                                                  | 3 (1.1)                                       | 0                       | 3 (1.1)                    | 6 (0.8)            |
| Hyperkalemia‡                                                                                            | 25 (9.3)                                      | 30 (11.4)               | 10 (3.8)                   | 65 (8.1)           |
| Safety assessments                                                                                       |                                               |                         |                            |                    |
| >30% Decline in eGFR from base-line to 30 days§                                                          | 17 (6.3)                                      | 10 (3.8)                | 3 (1.1)                    | 30 (3.8)           |
| Serum potassium level — no./total no. (%)¶                                                               |                                               |                         |                            |                    |
| >5.5 mmol/liter                                                                                          | 40/262 (15.3)                                 | 48/258 (18.6)           | 25/257 (9.7)               | 113/777 (14.5)     |
| >5.5 to ≤6.0 mmol/liter                                                                                  | 34/262 (13.0)                                 | 43/258 (16.7)           | 21/257 (8.2)               | 98/777 (12.6)      |
| >6.0 mmol/liter                                                                                          | 12/263 (4.6)                                  | 12/262 (4.6)            | 7/262 (2.7)                | 31/787 (3.9)       |

- \* The safety analysis population comprised all the participants who had undergone randomization and received at least one dose of a trial drug.
- † The participants with an adverse event reported here were those who had received at least one dose of a trial drug and had an adverse event that had started or worsened after the first dose and up to 3 days after any temporary or permanent interruption of the trial treatment.
- ‡ In the assessment of hyperkalemia, the investigators used the preferred terms “hyperkalemia” and “blood potassium increased” in the *Medical Dictionary for Regulatory Activities*, version 27.1.
- § The eGFR was calculated with the use of the Chronic Kidney Disease Epidemiology Collaboration equation,<sup>8</sup> which was modified for the Japanese participants.<sup>9</sup>
- ¶ The denominator represents all the participants at risk for an abnormal laboratory result. Participants must have had both a baseline and postbaseline (after the first dose and up to 3 days after any temporary or permanent interruption of the trial treatment) value, with the baseline value not exceeding the displayed threshold. The numerator represents the number of participants at risk who had at least one postbaseline laboratory assessment that met the criterion.

Table 3. Investigator-reported adverse events and laboratory safety assessments.

## Secondary Safety and Adverse Outcomes

**9.3%**

hyperkalemia in combo

**11.4%**

hyperkalemia in finerenone

**6.3%**

combo eGFR decline >30%

**7.1%**

combo serious adverse events

- Hyperkalemia: Combination therapy had a lower incidence of hyperkalemia (9.3%) than finerenone alone (11.4%), confirming the potassium-lowering synergy of SGLT2i.
- eGFR Decline: Acute eGFR decline >30% at day 30 was higher in combination (6.3%) than finerenone alone (3.8%) or empagliflozin alone (1.1%).
- Reversibility: The initial eGFR decline was reversible. Both eGFR and UACR returned toward baseline 30 days after stopping the drug (washout period).
- Serious Adverse Events: SAE rates were low and balanced across groups (7.1% in combination, 6.1% in finerenone, 6.4% in empagliflozin).

# Results Critical Appraisal

- Primary Outcome: Relative change in mean UACR at Day 180 was 29% greater in combination vs finerenone alone (ratio 0.71; 95% CI 0.61–0.82;  $P < 0.001$ ) and 32% greater vs empagliflozin alone (ratio 0.68; 95% CI 0.59–0.79;  $P < 0.001$ ).
- Additive Effect: UACR decreased by 52% from baseline in the combination group, matching the predicted additive effect of SGLT2i (29%) and finerenone (32%) monotherapies.
- Efficacy Thresholds: Higher proportions achieved  $>30\%$  UACR reduction (70.0% in combination vs 52.1% in finerenone vs 51.7% in empagliflozin; OR combo vs empa = 2.21).
- Washout Efficacy: Stopping therapy led to a 1.63-fold increase in UACR in the combination arm at Day 210, proving ongoing active treatment efficacy.

***Efficacy Verdict: The 52% UACR reduction in the combination arm reflects a robust, additive pharmacodynamic benefit that is highly statistically significant and clinically meaningful.***

# Risk of Bias 2 (RoB 2) Summary

|                                                                                                                                                                                 |                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| <b>Domain 1: Bias arising from the randomisation process</b><br>Low Risk (Interactive Web-response randomisation with stratification; baseline characteristics highly balanced) | LOW RISK                |
| <b>Domain 2: Bias due to deviations from intended interventions</b><br>Low Risk (Strict double-blind design with double placebos; protocol deviations were minor and balanced)  | LOW RISK                |
| <b>Domain 3: Bias due to missing outcome data</b><br>Low Risk (Low attrition rate; multiple imputation and tipping point analyses conducted)                                    | LOW RISK                |
| <b>Domain 4: Bias in measurement of the outcome</b><br>Low Risk (Urinary albumin and creatinine measured at central laboratory; assessors fully blinded)                        | LOW RISK                |
| <b>Domain 5: Bias in selection of the reported result</b><br>Low Risk (Trial prospectively registered, analysis plan pre-specified, all outcomes reported as planned)           | LOW RISK                |
| <b>OVERALL RISK OF BIAS JUDGEMENT</b>                                                                                                                                           | <b>LOW RISK OF BIAS</b> |

# GRADE Certainty of Evidence

## GRADE FINAL CERTAINTY

# HIGH CERTAINTY

*We have high confidence that the true effect lies close to that of the estimate of the effect. Further research is unlikely to change our confidence.*

**Outcome: Urinary albumin-to-creatinine ratio (UACR) reduction at 6 months**

### **Risk of Bias: NONE**

No downgrade. Multicenter, double-blind, randomized trial with low risk of bias.

### **Inconsistency: NONE**

No downgrade. Results consistent across subgroups (including age, eGFR, and baseline UACR).

### **Indirectness: NONE**

No downgrade. UACR reduction is a validated surrogate biomarker for kidney disease progression.

### **Imprecision: NONE**

No downgrade. Sample size (800) sufficient; 95% confidence intervals are narrow and exclude the null.

# DISCUSSION

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Key Trials, Guidelines, and Bayesian Reanalysis

# Chronology of Key Trials in CKD & Diabetes

| Study (Year)       | Population                                                 | Intervention                               | Key Findings & Clinical Relevance                                                                                                      |
|--------------------|------------------------------------------------------------|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| CREDENCE (2019)    | CKD & T2D on RASi, eGFR 30-90, UACR 300-5000 (n=4,401)     | Canagliflozin vs. Placebo                  | Reduced composite kidney failure & CV death by 30%. Established SGLT2 inhibitors as primary organ-protective standard of care.         |
| DAPA-CKD (2020)    | CKD with/without T2D, eGFR 25-75, UACR 200-5000 (n=4,304)  | Dapagliflozin vs. Placebo                  | Reduced composite renal decline & CV death by 39%. Confirmed SGLT2i benefits extend to non-diabetic CKD.                               |
| FIDELIO-DKD (2020) | CKD & T2D, eGFR 25-75, UACR 30-5000 (n=5,674)              | Finerenone vs. Placebo                     | Reduced composite renal events (ESKD, $\geq 40\%$ eGFR drop) by 18%. Established finerenone as a key add-on therapy.                   |
| FIGARO-DKD (2021)  | Broader CKD & T2D, eGFR $\geq 25$ , UACR 30-5000 (n=7,352) | Finerenone vs. Placebo                     | Reduced composite CV events by 13%, driven by heart failure hospitalizations. Confirmed CV benefits of finerenone.                     |
| CONFIDENCE (2025)  | CKD & T2D on RASi, eGFR 30-90, UACR 100-5000 (n=800)       | Simultaneous Combination vs. Monotherapies | First trial of simultaneous initiation. Shows 52% additive UACR reduction. Combination reduces hyperkalemia risk vs. finerenone alone. |

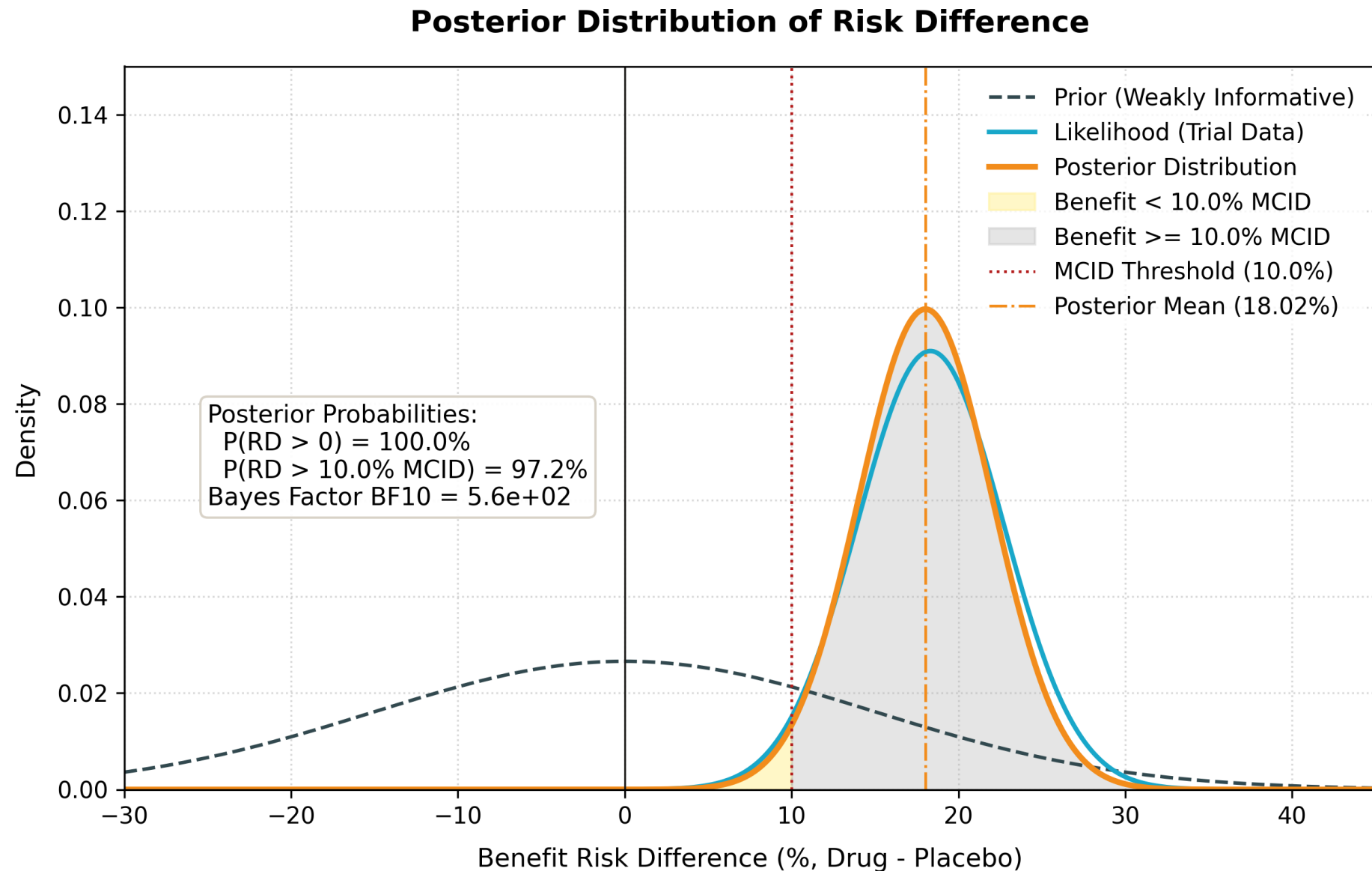
# Current Guideline Consensus on Combined Therapy

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- KDIGO 2024 Clinical Practice Guideline: Recommends first-line therapy with an ACEi/ARB and SGLT2i. Finerenone (nonsteroidal MRA) is recommended as add-on for patients with persistent albuminuria.
- ADA Standards of Care in Diabetes (2026): Supports using both SGLT2 inhibitors and finerenone for independent cardiovascular and kidney organ protection in type 2 diabetes.
- Overcoming Clinical Inertia: In practice, stepwise drug addition introduces therapeutic lag and increases follow-up loss. Simultaneous initiation offers immediate dual-pathway protection.
- Consensus Verdict: Simultaneous dual initiation of finerenone and empagliflozin is physiologically rational, highly effective, and safe under appropriate clinical monitoring.

# Posterior Distribution of Risk Difference (Weakly Informative Prior)

Conjugate prior/likelihood/posterior distribution for the risk difference scale for binary endpoints (MCID=10.0%)



# Posterior Efficacy under Alternative Priors

Sensitivity analysis of posterior probabilities under different prior assumptions

| Prior Profile                | Prior Mean (SD) * | Posterior Mean RD (95% CrI) | P(RD > 0) | P(RD > 10.0% MCID) |
|------------------------------|-------------------|-----------------------------|-----------|--------------------|
| Weakly Informative (Default) | 0.00 (2.0000)     | 18.02% (9.76% to 25.42%)    | 100.0%    | 97.2%              |
| Skeptical (MCID-derived)     | 0.00 (0.3190)     | 13.72% (6.19% to 20.69%)    | 100.0%    | 84.2%              |
| Enthusiastic (MCID-derived)  | 0.41 (0.7795)     | 17.73% (9.66% to 24.99%)    | 100.0%    | 97.0%              |
| Historical (Phase 2 Data)    | 0.80 (0.4000)     | 18.27% (10.83% to 25.00%)   | 100.0%    | 98.5%              |

\* Note: Prior mean and SD are specified on the log-odds-ratio scale. Risk differences (RD) are drug minus placebo (positive values favor treatment). MCID is 10.0% absolute risk difference.

# APPLICABILITY & TAKEAWAYS

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Local Settings, Ward Guidelines, and take-aways

# Local Applicability: North Indian / Ghaziabad Patients

## CLINICAL & STEWARDSHIP BENEFITS

- High Albuminuria Prevalence: A large proportion of diabetic patients visiting Santosh Medical College Ghaziabad present with advanced CKD and albuminuria, where dual therapy is highly indicated.
- Clinical Inertia Reduction: Outpatient workloads are extremely high; simultaneous dual initiation prevents therapeutic lag from stepwise outpatient titration.
- Potassium-balancing Safety: Dual therapy reduces the risk of hyperkalemia compared to finerenone alone, which is a major safety advantage in settings with limited lab access.

## FEASIBILITY & MONITORING CAVEATS

- Financial Barriers: Empagliflozin is widely available in India in generic formats, but finerenone is branded and represents a substantial monthly out-of-pocket cost.
- Initial eGFR Dip: The reversible initial eGFR dip (6.3% of patients had >30% decline) requires outpatient education to avoid premature, fear-based discontinuation.

***Ghaziabad Clinical Context: Simultaneous dual initiation reduces therapeutic lag and outpatient follow-up loss, but requires addressing financial barriers and patient education on sick-day rules.***

# Clinical Takeaways for OPD, Wards, & ICU/HDU

## OUTPATIENT CLINICS (OPD)

- Consider simultaneous dual initiation in stable CKD + T2D patients with persistent albuminuria (UACR >100 mg/g) who are already on max tolerated RASi.
- Labs Protocol: Check baseline potassium and eGFR; repeat at 2-4 weeks, then every 3-6 months.
- Patient Education: Reinforce the safety of the initial reversible eGFR dip and ensure follow-up compliance.

## INPATIENT WARDS

- Acute Illness Rules: Temporarily hold SGLT2i and finerenone during acute gastroenteritis, high fever, sepsis, or acute kidney injury to avoid hypotension and ketoacidosis.
- Discharge Planning: In stable patients admitted for non-renal causes, review and optimize organ-protective therapies (RASi, SGLT2i, MRA) before discharge.
- Transition: Ensure clear discharge counseling on sick-day rules.

## ICU & HIGH DEPENDENCY (HDU)

- Strict Hold: Do not initiate or continue SGLT2i and finerenone in hemodynamically unstable patients, shock, or severe sepsis.
- Ketoacidosis Risk: SGLT2i must be avoided in patients who are fasting, post-operative, or critically ill due to the risk of euglycemic diabetic ketoacidosis (DKA).
- Stabilization: Re-evaluate and re-initiate only after full recovery and return to normal oral intake.

# Journal Club Discussion Questions

## CLINICAL PRACTICE

1. Under what clinical circumstances would you prefer simultaneous dual initiation of finerenone and empagliflozin over a traditional step-up approach?

## SURROGATE OUTCOMES

2. Does a 52% reduction in UACR (a surrogate endpoint) over 6 months justify the immediate initiation of dual therapy before long-term hard outcome data are published?

## FINANCIAL BARRIERS

3. How can we navigate the financial barriers of branded finerenone for patients in Ghaziabad while still providing evidence-based organ protection?

## MONITORING LOGISTICS

4. What practical strategies can we implement in our medicine OPD to ensure timely potassium and eGFR checks for patients starting dual therapy?

# References

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- Agarwal R, Green JB, Heerspink HJL, et al. Finerenone with Empagliflozin in Chronic Kidney Disease and Type 2 Diabetes. N Engl J Med 2025;393:533-543.
- The CREDENCE Trial Investigators. Canagliflozin and Renal Outcomes in Type 2 Diabetes and Nephropathy. N Engl J Med 2019;380:2295-2306.
- The DAPA-CKD Trial Investigators. Dapagliflozin in Patients with Chronic Kidney Disease. N Engl J Med 2020;383:1436-1446.
- The FIDELIO-DKD Trial Investigators. Effect of Finerenone on Chronic Kidney Disease Outcomes in Type 2 Diabetes. N Engl J Med 2020;383:1895-1907.
- The FIGARO-DKD Trial Investigators. Cardiovascular Events with Finerenone in Kidney Disease and Type 2 Diabetes. N Engl J Med 2021;385:2152-2163.